# Network Science

## Lecture 03: Strong and Weak Ties

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# Strong and Weak Ties

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## Focus

- triadic closure and strong triadic closure
- bridges and local bridges
- why weak ties matter for global connectivity
- empirical evidence from phone and social-media data

# Module 1

## Triadic Closure

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$$\text{shared friend} \Rightarrow \text{increased probability of closure}$$

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An **open triad** has two edges and one missing edge. **Triadic closure** is the process by which the missing edge appears over time.

> *Consider three nodes A, B, C where A–B and A–C exist but B–C does not. Triadic closure predicts B–C will form.*

Ask: before we explain why, what mechanisms could drive the missing edge to appear?

- Shared friends create environments to meet.
- Trust can be borrowed through a common contact.
- Social friction rewards closing the triangle to resolve inconsistency.

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Ask: before we explain why, what mechanisms could drive the missing edge to appear?

- **Opportunity:** Shared friends create environments to meet.
- **Trust:** Trust can be borrowed through a common contact.
- **Incentive:** Social friction rewards closing the triangle to resolve inconsistency.

# Strong Triadic Closure

> *If A has **strong** ties to both B and C, and B–C does not exist, then A violates the Strong Triadic Closure Property.*

**Strong Triadic Closure Property:** If a node maintains strong ties to two distinct neighbors, there must be at least a weak tie between those neighbors.

# Measuring Closure: Clustering Coefficient

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where $k_i$ is the degree of node $i$. Values range from 0 (no triangle) to 1 (complete neighborhood).

# Quick Check 3.A: Clustering Coefficient

Node $B$ has 4 neighbors: $A, C, D, E$. Among $B$'s neighbors, exactly 2 edges exist: $(A, C)$ and $(C, D)$.

1. Compute $C_B$.
2. Is $B$'s neighborhood highly clustered or sparse? What would that imply about $B$'s social position?

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2. **Interpretation:**

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**2. Interpretation:**

$C_B = 1/3$ indicates a sparse but not isolated neighborhood. $B$ connects nodes that do not all know each other — a position with potential brokerage value, since $B$ mediates between sub-clusters $A, C$ and $D$.

# Takeaway

> Triadic closure is a local mechanism, but repeated across a network it explains why social graphs become clustered — and why those clusters can be structurally separated by rare long-range ties.

# Module 2

## Bridges and Local Bridges

> **Guiding Question:** Which edges matter most for keeping information flowing across a network?

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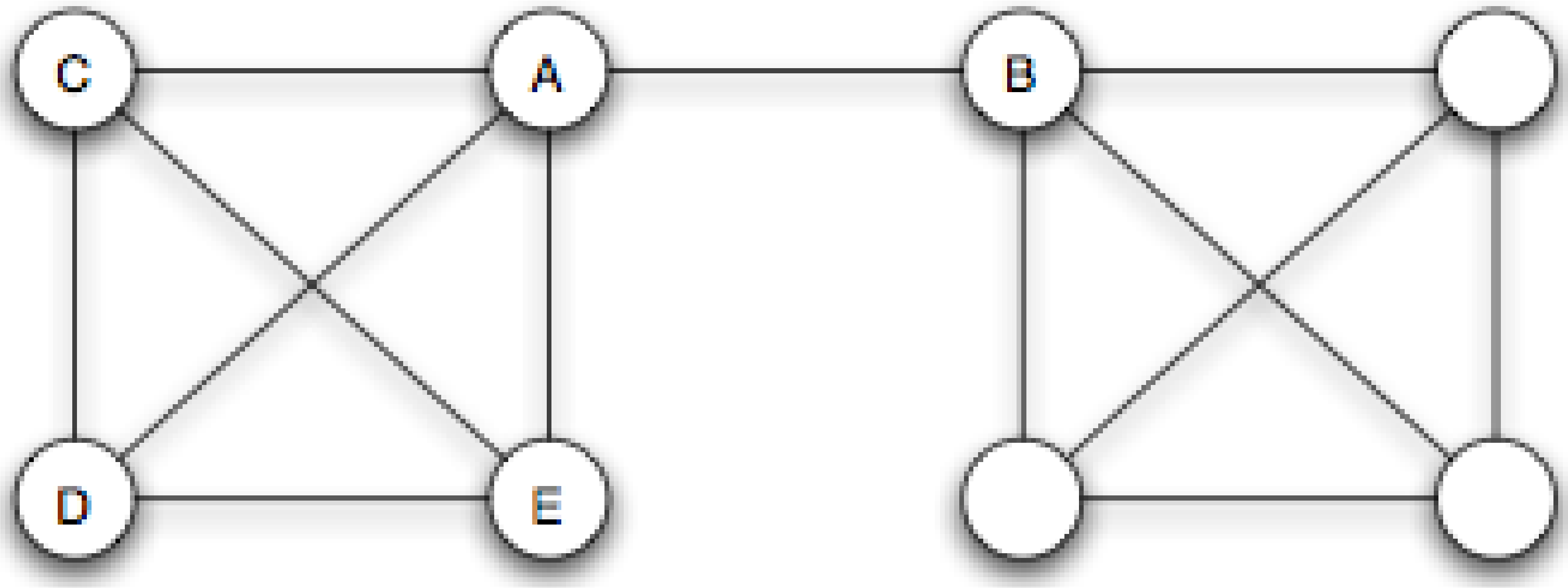
$$\text{important edge} \neq \text{strongest edge}$$

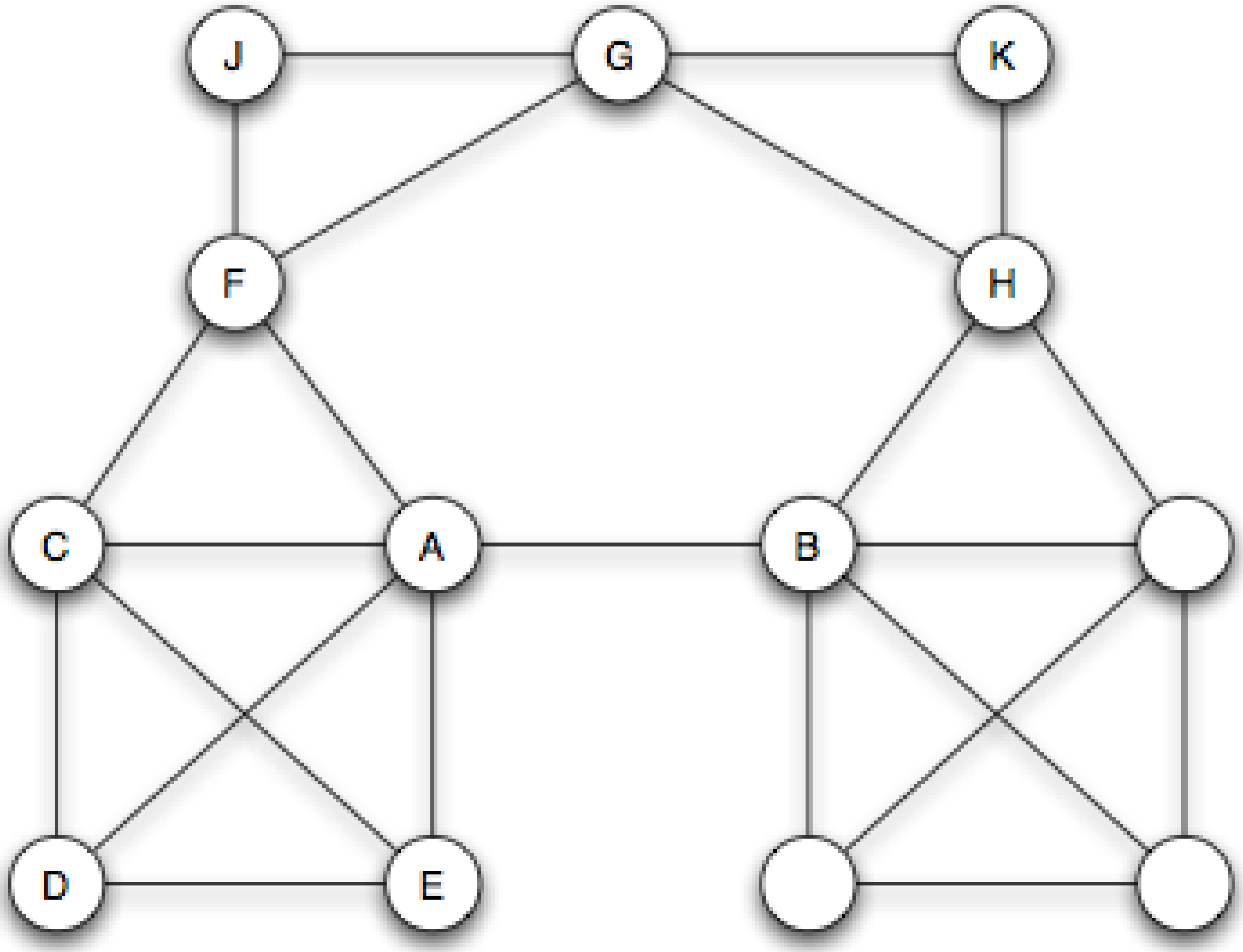
# Bridge




Source: Easley & Kleinberg 2010

A **bridge** is an edge whose removal increases the number of connected components. Equivalently: a bridge lies on no cycle.

# Local Bridge

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A **local bridge** links two nodes that share **no** neighbors. Removing it increases the distance between its endpoints to more than two.

# Structural Holes and Social Capital

A node that acts as the sole link between otherwise disconnected groups spans a **structural hole** (Burt, 1992).

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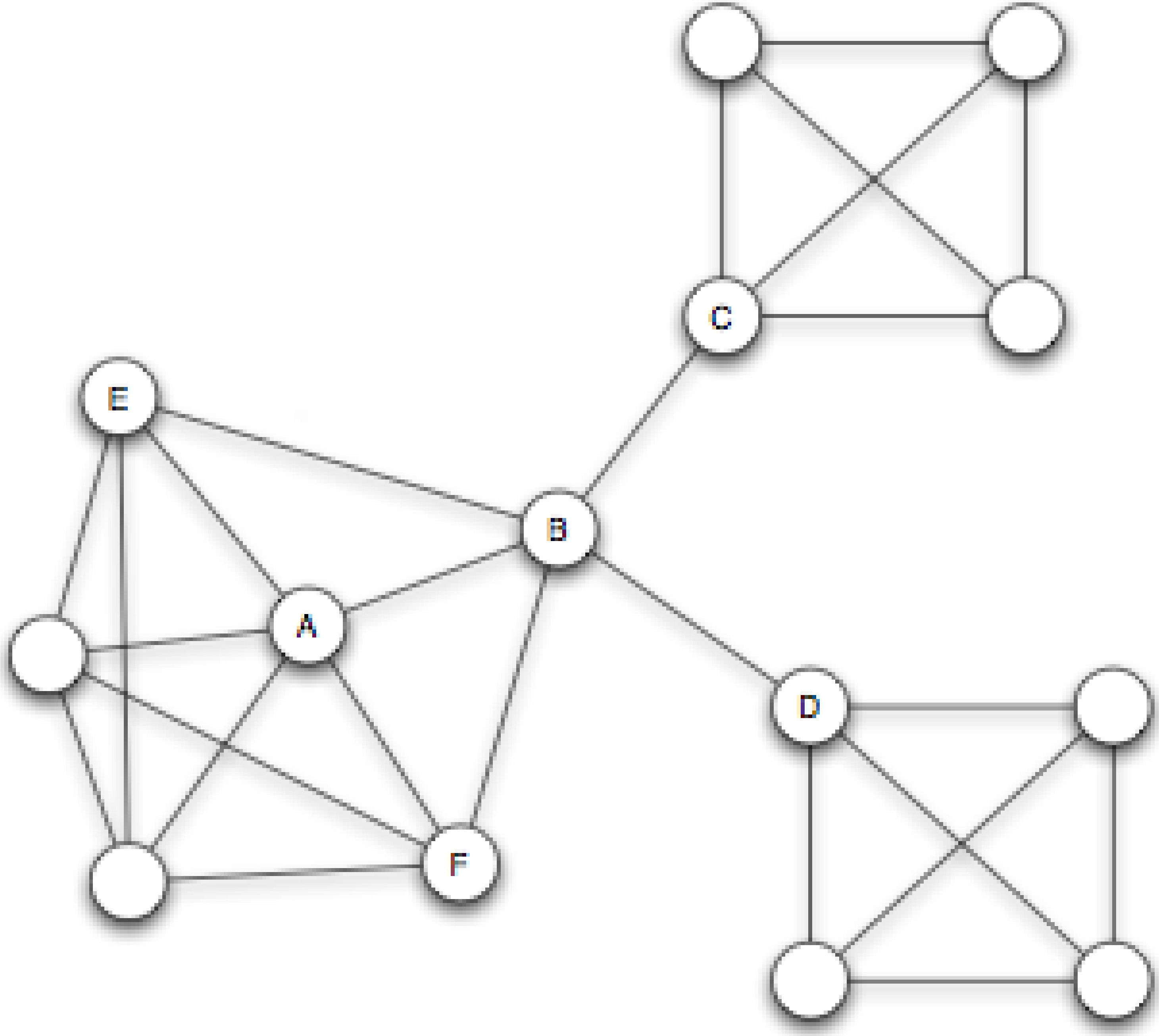


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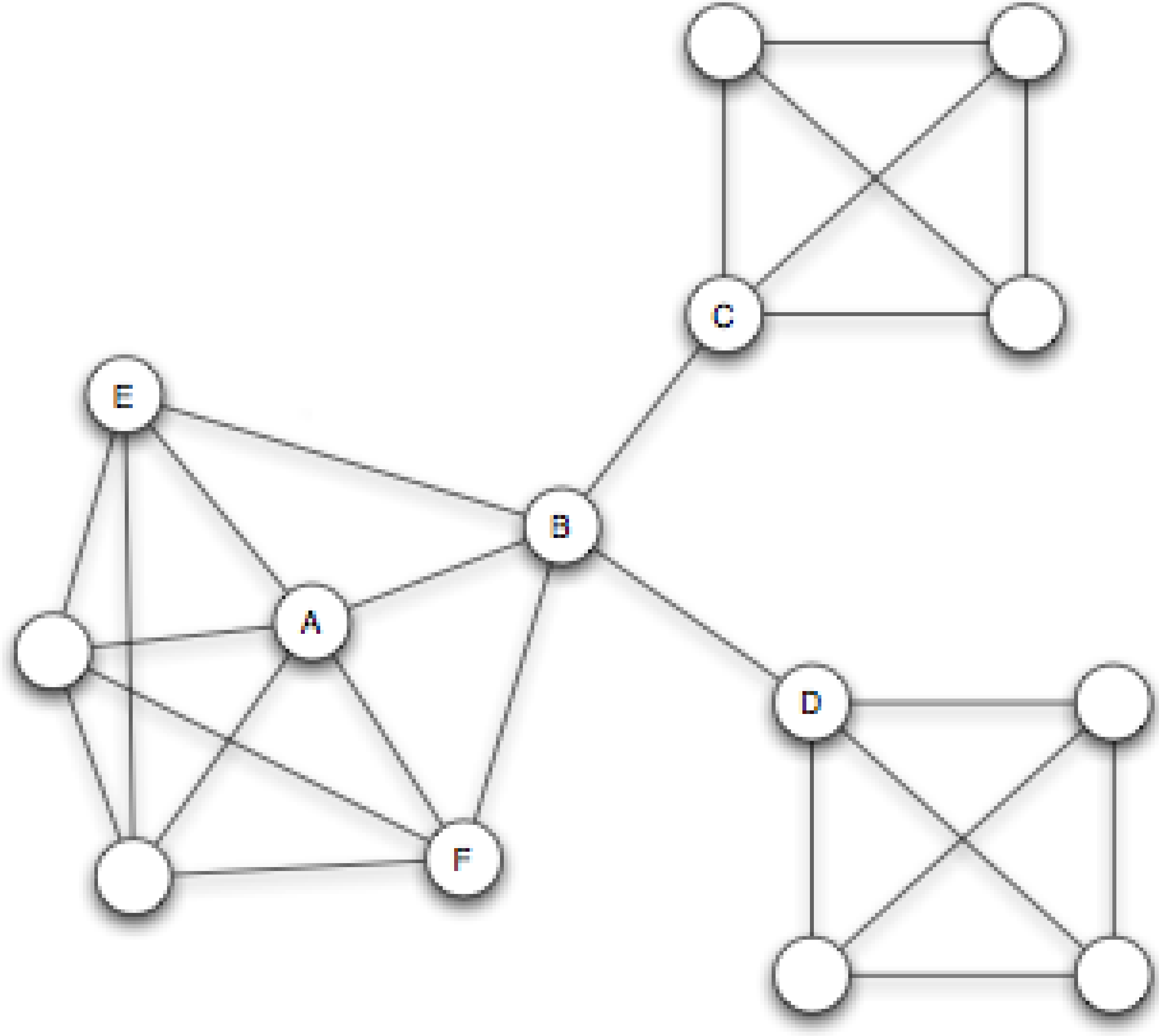


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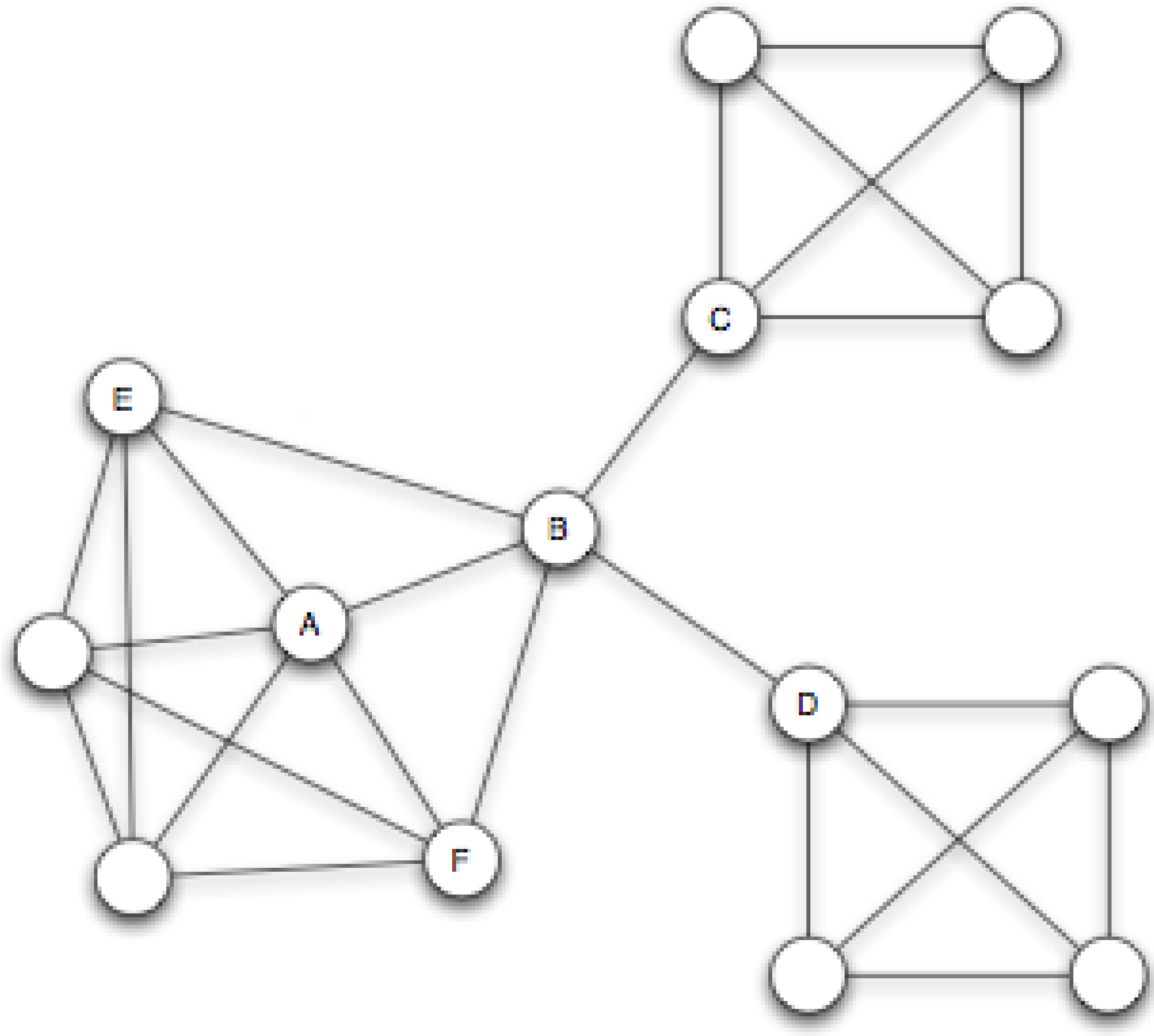


Source: Burt 1992

# Quick Check 3.B: Bridges and Brokers

The graph below represents a small research community. Each letter is a researcher.

> *A — B — C — D — E, plus edges B–C and C–D forming triangles.*

1. Which edges in this graph are **true bridges**?
2. Your advisor says "C is a broker". Do you agree? Justify using the definitions of local bridge and structural hole.

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# Quick Check 3.B: Solution

**1. True bridges:** Edges **A–B** and are bridges — removing either disconnects a node from the rest of the graph. Edges B–C and C–D lie on cycles and are not bridges.

No. C sits inside the triangle B–C–D, meaning B and D are already connected. C spans no structural hole: all of C's neighbors already know each other. A true broker would connect groups with no direct path between them.

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# Quick Check 3.B: Solution

**1. True bridges:** Edges **A–B** and **D–E** are bridges — removing either disconnects a node from the rest of the graph. Edges B–C and C–D lie on cycles and are not bridges.

**2. Is C a broker?** No. C sits inside the triangle B–C–D, meaning B and D are already connected. C spans no structural hole: all of C's neighbors already know each other. A true broker would connect groups with no direct path between them.

# Takeaway

> Structurally crucial edges are identified by position, not by weight. A node spanning structural holes gains information and control advantages — but loses them the moment triadic closure reconnects its contacts.

# Module 3

## Weak Ties as Structural Connectors

> **Guiding Question:** Why should local bridges tend to be weak ties rather than strong ties?

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**Guiding Question:** Why should local bridges tend to be weak ties rather than strong ties?

$$\text{local bridge} + \text{strong triadic closure} \Rightarrow \text{weak tie}$$

# Theorem Statement

**Under strong triadic closure, every local bridge must be a weak tie.**

# Why Might This Be True?

Suppose A–D was a **strong** tie and also a local bridge. Then by strong triadic closure, A's other strong neighbors (B, C) must each have a tie to D. But then D shares neighbors with A — contradicting the local-bridge condition.

> *The assumption "A–D is strong AND a local bridge" forces a neighbor of D to exist among A's friends, destroying the local-bridge condition.*

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# Granovetter: "Getting a Job" (1973)

Granovetter interviewed Boston-area residents who had recently found a new job through a social contact.

Is it the people we see most often who provide the most useful job leads?

Most job leads came from acquaintances seen rarely — not from close friends. The contacts providing novel opportunities were on the social periphery, not the social core.

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**Result:** Most job leads came from acquaintances seen rarely — not from close friends. The contacts providing novel opportunities were on the social periphery, not the social core.

# Interpretation

The theorem links a local social concept (tie strength) to a global structural one (bridging separate communities).

Weak ties are the primary channels through which non-redundant information — new jobs, new ideas, different perspectives — reaches individuals across community boundaries.

# Quick Check 3.C: Strong Ties and Local Bridges

A network has three nodes: A, B, C. A–B is a **strong** tie. A–C is a **strong** tie. There is no edge between B and C.

1. Does this network satisfy the **Strong Triadic Closure Property**?
2. Is the edge A–B a local bridge? Can you tell from the information given?

# Quick Check 3.C: Solution

No. A has strong ties to B and C, but B and C are not connected — this violates STC.

**2. Local bridge status of A–B:** In this three-node network, A and B share no common neighbors (C connects to A but not to B). So . But the Weak Tie Theorem does not apply here — its precondition (STC holds) is violated. The theorem makes no claim about what happens when STC fails.

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**2. Local bridge status of A–B:** In this three-node network, A and B share no common neighbors (C connects to A but not to B). So **A–B is a local bridge in this graph**. But the Weak Tie Theorem does not apply here — its precondition (STC holds) is violated. The theorem makes no claim about what happens when STC fails.

# Takeaway

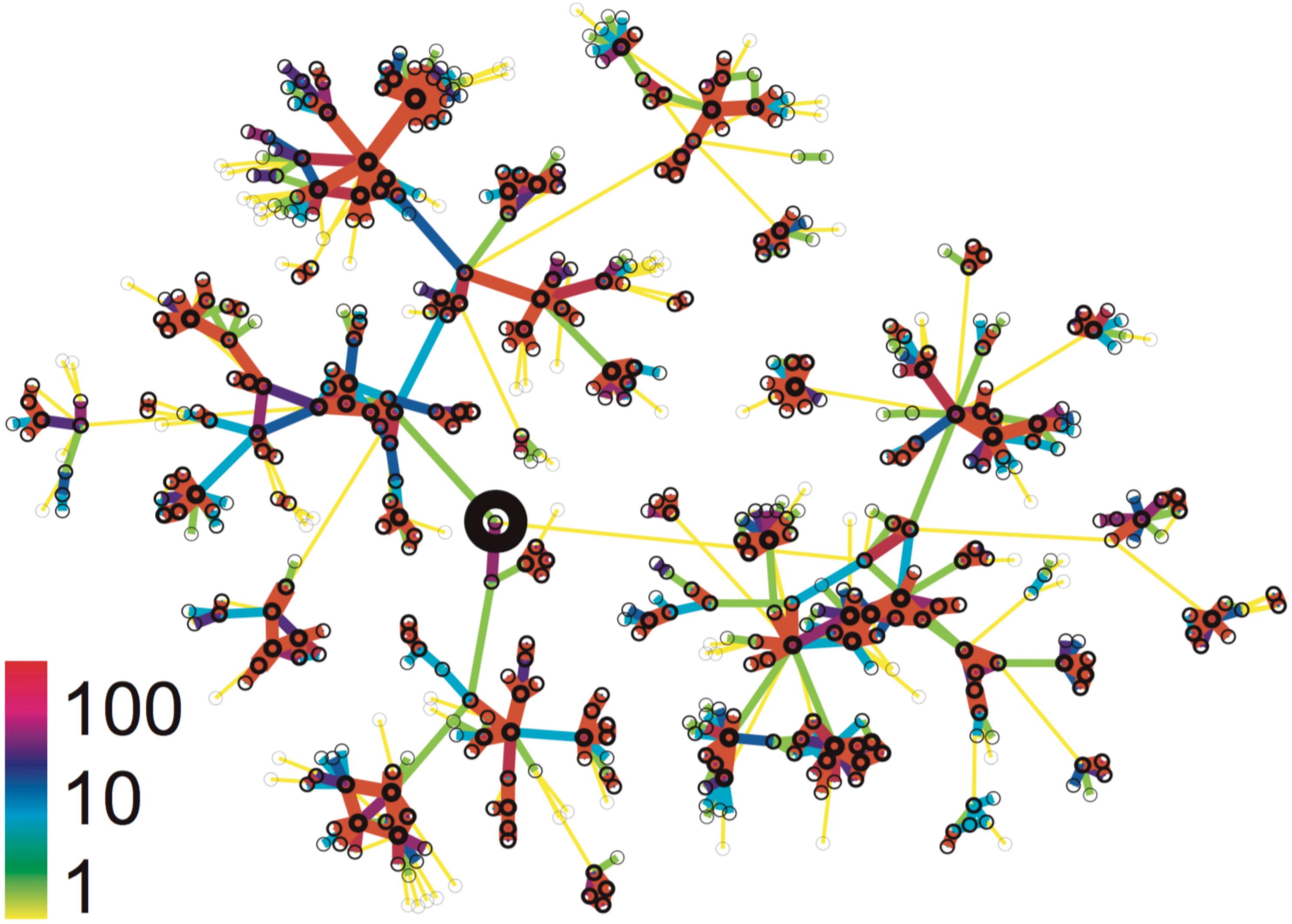
> Weak ties are structurally positioned as bridges between communities. The theorem explains *why* — not by sociology, but by graph structure: strong ties cluster locally, and local clustering destroys the bridge condition.

# Module 4

## Cell-Phone Evidence

> **Guiding Question:** Does the weak-ties theorem survive contact with real-world data at scale?

# Setup: Modeling Communication as a Graph




*Source: Onnela et al. 2007*

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Onnela et al. (2007) modeled nationwide cell-phone call logs as an undirected weighted graph. Edge weight = total call duration between two numbers.

# Formalizing Almost-Local Bridges

In a real weighted graph, true local bridges are rare. Instead, we measure **Neighborhood Overlap**:

$$O(u, v) = \frac{|N(u) \cap N(v)|}{|N(u) \cup N(v)|}$$

where $N(u)$ is the set of neighbors of $u$, excluding $v$ (and vice versa).

What does $O(u, v) = 0$ mean?

The two endpoints share common neighbors — the edge is a perfect local bridge.

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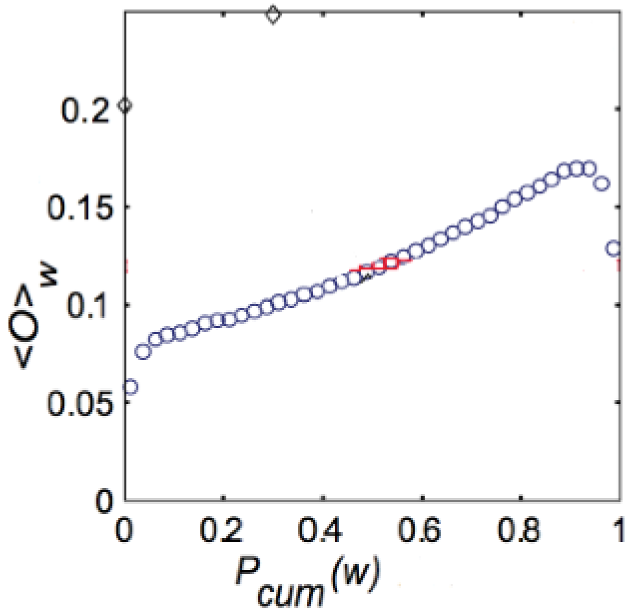
# Evidence: Tie Strength vs. Overlap

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Does tie strength predict neighborhood overlap?

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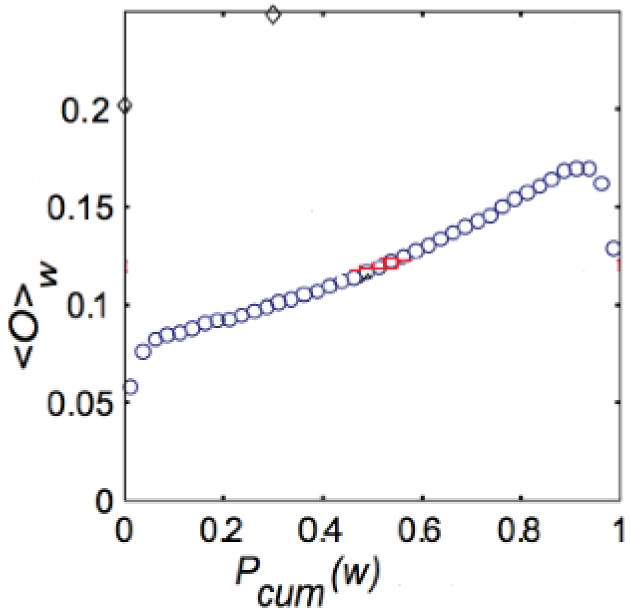


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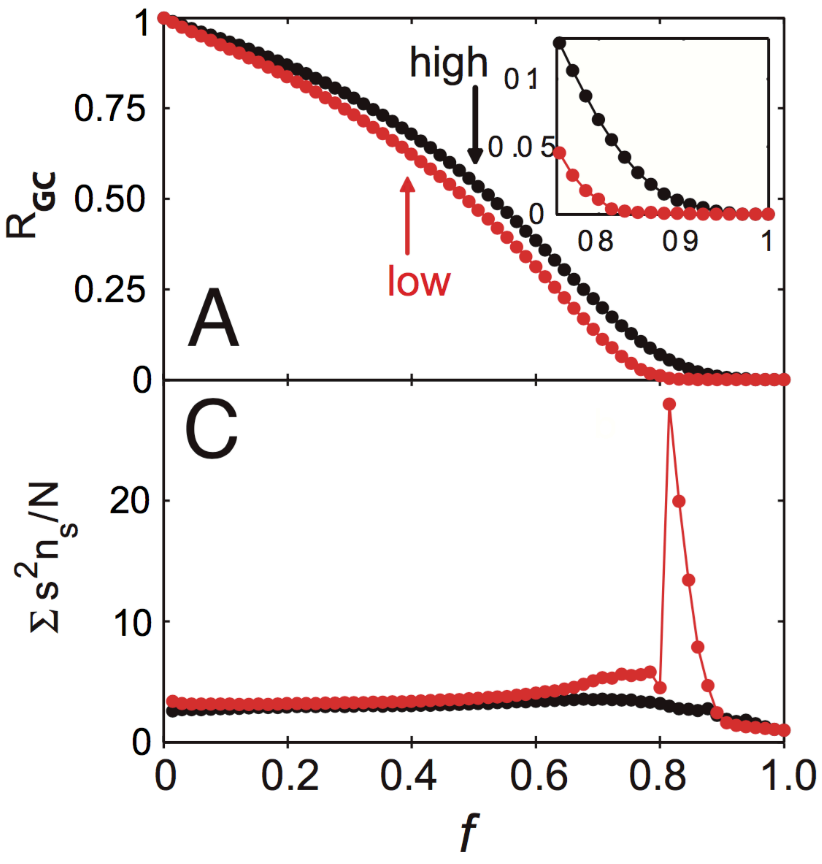
Does tie strength predict neighborhood overlap?
Yes: weak ties (low strength percentile) cluster near

$$O \approx 0$$

. Strong ties accumulate at higher overlap values — consistent with Granovetter's prediction.

# Evidence: Network Robustness

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Which edge removal strategy breaks the network fastest?

Removing edges in

tie-strength order (weakest first) fragments the giant component far faster than removing strongest ties first.

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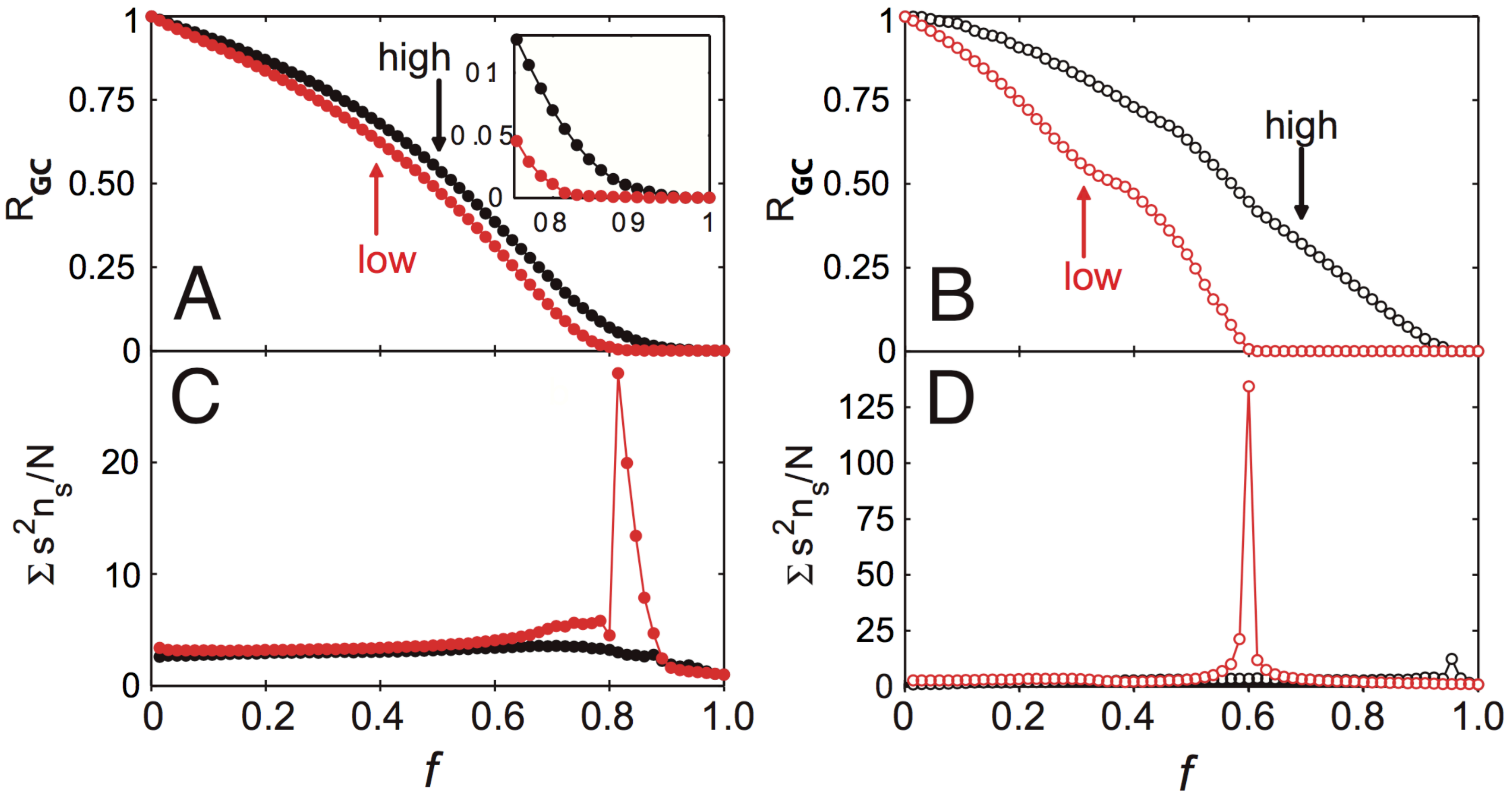


Source: Onnela et al. 2007

Which edge removal strategy breaks the network fastest?

Removing edges in **increasing** tie-strength order (weakest first) fragments the giant component far faster than removing strongest ties first.

# Quick Check 3.D: Neighborhood Overlap

Nodes $A$ and $B$ are connected. Their neighbor sets (excluding each other) are:

- $N(A) = C, D, E$
- $N(B) = D, E, F$

1. Compute $O(A, B)$.
2. Is A–B closer to a local bridge or a clique edge? What does this imply about the information A would receive through B?

# Quick Check 3.D: Solution

A–B has moderate overlap. It is not a local bridge ($O > 0$), but nor is it embedded in a clique ($O < 1$). A receives partly redundant information through B. If $O$ were near 0, B would be A's primary source of information from a distant community.

# Quick Check 3.D: Solution

$$O(A, B) = \frac{|D, E|}{|C, D, E, F|} = \frac{2}{4} = 0.5$$

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**Interpretation:** A–B has moderate overlap. It is not a local bridge ($O > 0$), but nor is it embedded in a clique ($O < 1$). A receives partly redundant information through B. If $O$ were near 0, B would be A's primary source of information from a distant community.

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# Takeaway

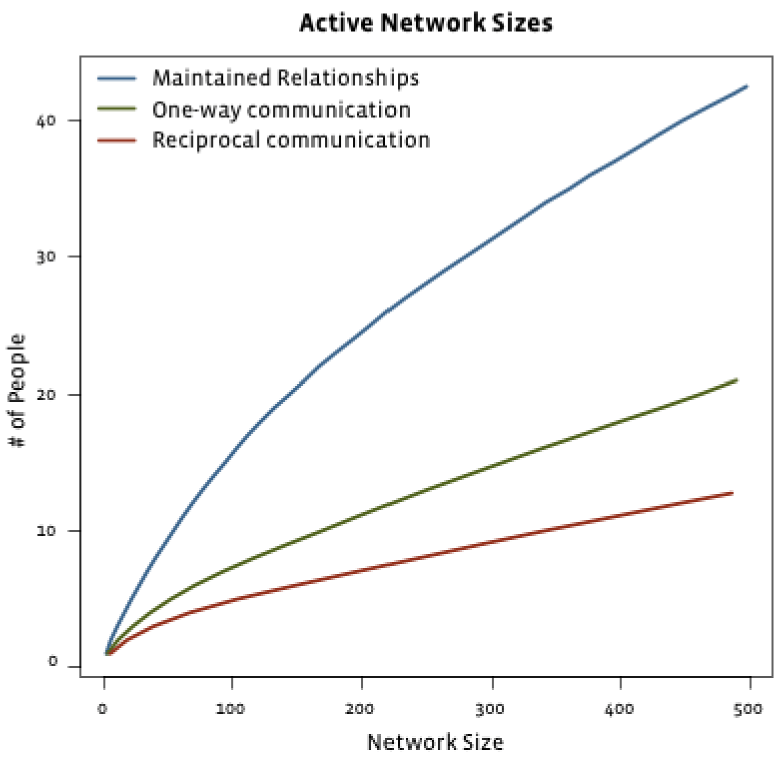
The weak-ties argument is not sociological speculation — it has measurable structural signatures at scale. Neighborhood overlap operationalizes the bridge concept, and empirical data confirm that weak ties carry it.

# Module 5

## Social Media

> **Guiding Question:** How do strong and weak ties change when the network moves to online platforms

# Evidence: Facebook (2009)




*Source: Easley & Kleinberg 2010*

As total friend count grows, what happens to actively maintained relationships?

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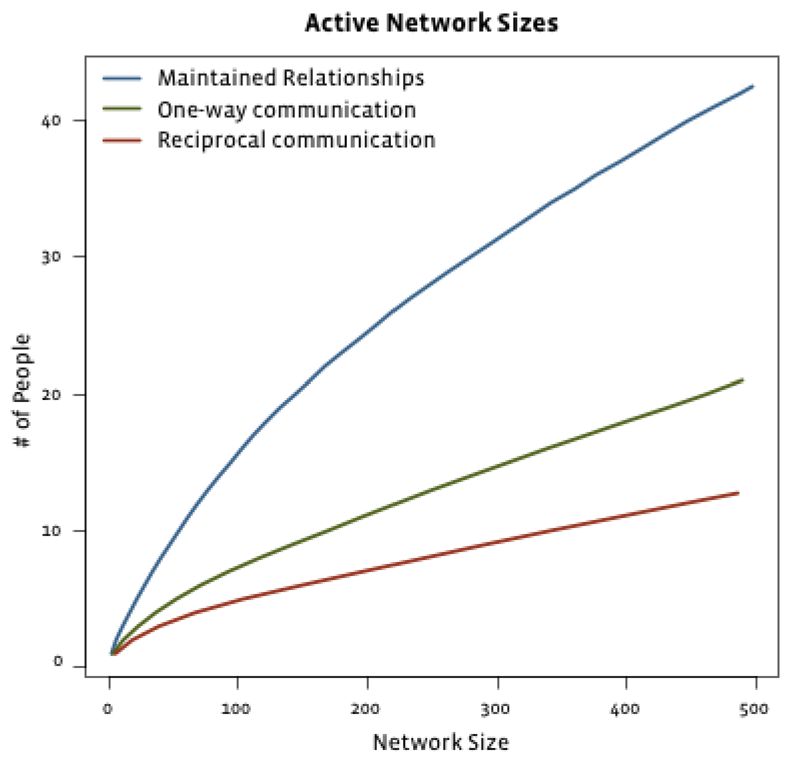
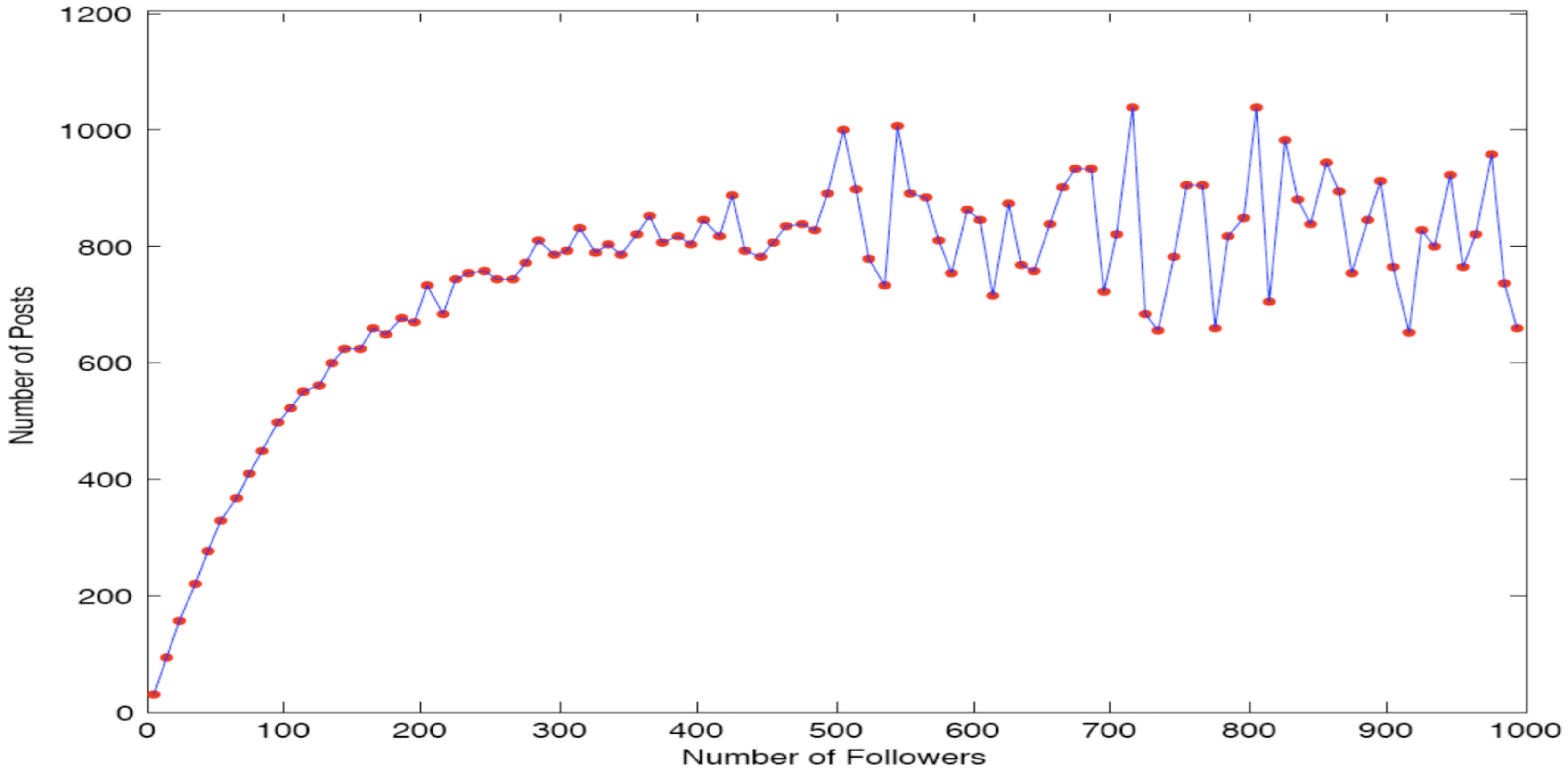



*Source: Easley & Kleinberg 2010*

As total friend count grows, what happens to actively maintained relationships?

Only a small, roughly constant number are "maintained" (reciprocal communication). The rest are passive weak ties.

# Evidence: Twitter (Huberman et al.)

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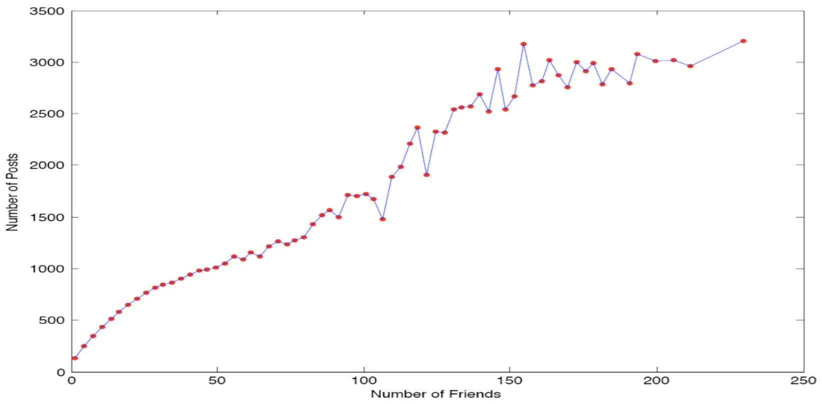


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On asymmetric platforms, the *follower* count can reach millions (zero-cost weak ties), while *friends* — mutual contacts with whom one actually interacts — remain a small subset.

# Interpretation

Does social media collapse the distinction between weak and strong ties?

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Does social media collapse the distinction between weak and strong ties?

No — it amplifies it. Platforms remove the cost of maintaining weak ties (passive contact, one-click follow), but strong ties still require active, reciprocal effort. The result is a much wider weak-tie periphery surrounding an unchanged strong-tie core.

# Quick Check 3.E: Dense vs. Sparse Follower Networks

Two students, X and Y, both use the same online platform. X follows 2000 accounts; Y follows 80 accounts but everyone follows everyone else.

1. Who is likely to have a higher average neighborhood overlap across their edges?
2. What does this imply about the kind of information X and Y each receive?

# Quick Check 3.E: Solution

Y likely has average overlap. Y's 80 contacts form a dense mutual cluster — many shared neighbors per edge. X's 2000 contacts include many algorithm-driven follows with near-zero shared neighborhood.

- X receives information — the structural signature of many weak ties and local bridges.
- Y receives information — the structural signature of a dense clique with few connections outside.

# Quick Check 3.E: Solution

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**2. Information implications:**

- X receives **diverse, non-redundant** information — the structural signature of many weak ties and local bridges.
- Y receives information — the structural signature of a dense clique with few connections outside.

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**2. Information implications:**

- X receives **diverse, non-redundant** information — the structural signature of many weak ties and local bridges.
- Y receives **redundant, reinforced** information — the structural signature of a dense clique with few connections outside.

# Takeaway

> Social media does not remove the weak/strong-tie distinction — it stretches the weak-tie periphery enormously while leaving the strong-tie core bounded by human cognitive limits.

# Wrap-Up

- local friendship structure creates predictable global effects
- weak ties are often not socially strongest, but structurally most valuable
- empirical network data broadly supports the theoretical pattern

## Looking Ahead: Lecture 04

How do we identify groups and roles within a network? Which nodes are important — and by what definition?

# Image Credits & References


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